Investigating Multimodal Polygonal Prompting: Translating Geometric Form via LLM-Augmented Generative Systems

Abstract

This report investigates the concept of Multimodal Polygonal Prompting, exploring the translation of abstract geometric properties of polygons—such as edge count, angular sharpness, symmetry, and complexity—into auditory, tactile, and olfactory sensory outputs. Central to this investigation is the role of Large Language Model (LLM)-augmented generative systems, which offer potential mechanisms for interpreting polygonal features and orchestrating cross-modal generation. Drawing upon research in cross-modal correspondences, computational geometry, generative AI, sensory representation ontologies, and human-computer interaction (HCI), this report analyzes the feasibility and challenges of such translations. Key findings highlight the necessity of explicit geometric feature extraction due to current limitations in multimodal LLM geometric perception, the potential for LLMs to leverage semantic mediation for bridging modalities (especially olfaction), and the possibility of AI discovering novel cross-modal relationships. A conceptual mapping framework and a system architecture proposal are presented, outlining strategies for translating specific polygonal features into corresponding sensory parameters. Evaluation metrics and inherent challenges, including perceptual subjectivity and technological limitations, are discussed. The report concludes by reflecting on the significance of this approach for applications in data representation, accessibility, generative art, and HCI, while identifying critical areas for future research.

1. Introduction

1.1. Defining Multimodal Polygonal Prompting

Multimodal Polygonal Prompting refers to the concept of utilizing polygonal shapes, defined by their geometric characteristics, as input prompts to generate corresponding outputs across multiple non-visual sensory modalities, specifically auditory, tactile, and olfactory domains. The core focus lies in translating the intrinsic *form* properties of these polygons—quantifiable attributes such as the number of edges (n), the degree of angularity or sharpness at vertices, the presence and type of symmetry, overall size or scale, and measures of geometric complexity or regularity—into analogous or corresponding sensory experiences. This approach moves beyond traditional text- or image-based prompting, exploring how abstract geometric information itself can serve as a control signal for generative systems.

1.2. The Role of LLM-Augmented Generative Systems

The realization of Multimodal Polygonal Prompting hinges on the capabilities of advanced Artificial Intelligence (AI), particularly the integration of Large Language Models (LLMs) with specialized generative models designed for different sensory modalities. LLMs possess powerful semantic understanding and generation capabilities, making them suitable candidates for interpreting the "meaning" encoded within a polygon's geometric features and translating this into instructions for other AI systems. While current multimodal generative model research often focuses on simulating the real world through visual, video, 3D, and 4D representations, aiming to capture appearance, dynamics, and geometry within these domains [1], the systematic integration and translation *across* disparate sensory modalities (vision-to-sound, vision-to-touch, vision-to-smell) based on abstract input like geometric form remains less explored. LLM-augmented systems offer a potential pathway by acting as a central "translator" or "orchestrator," processing the geometric input (potentially represented as structured data or descriptive text) and directing modality-specific generative models (e.g., text-to-audio, parameter-driven haptic renderers, descriptor-to-scent generators) to produce the desired sensory outputs.

1.3. Rationale and Significance

Investigating Multimodal Polygonal Prompting holds significant potential across various domains. It offers novel methods for **data representation and exploration**, allowing complex geometric data or relationships to be perceived through channels beyond vision. This has profound implications for **accessibility**, potentially enabling individuals with visual impairments to understand geometric concepts or interact with graphical information through sound and touch.[3] In **computational creativity and generative art**, it opens avenues for creating novel multisensory experiences derived from geometric forms. Within **Human-Computer Interaction (HCI)**, it could lead to new interaction paradigms where geometric shapes act as controls or feedback mechanisms. Furthermore, this research contributes to the fundamental understanding of **cross-modal cognition** by exploring how abstract features are mapped across senses, potentially revealing insights into the underlying principles of perception. It aligns with the broader technological goal of digitizing and generating experiences for senses beyond vision and audition, such as olfaction.[5]

1.4. Scope and Structure

This report provides a comprehensive technical analysis of Multimodal Polygonal Prompting. It focuses specifically on the translation of polygonal *form* features (edge count, angularity, symmetry, complexity, size) into auditory, tactile, and olfactory representations using LLM-augmented generative AI. It excludes the translation of non-form properties like color, internal texture, or pre-assigned semantic labels unless directly relevant to form interpretation. The report is structured as follows:

- **Section 2:** Examines the concept through five key analytical lenses: Cross-Modal Correspondences, Computational Geometry & Perceptual Shape Descriptors, Generative AI Capabilities, Sensory Representation Ontologies, and HCI/Interaction Design.
- **Section 3:** Proposes specific translation strategies and consolidates them into a Mapping Framework.
- **Section 4:** Outlines a conceptual System Architecture for an LLM-augmented generative system.
- **Section 5:** Discusses Evaluation Metrics and key challenges in assessing the success of such translations.
- **Section 6:** Offers Reflexive Insights on the approach, its implications, and future research directions.
- **Section 7:** Provides concluding remarks.

2. Analytical Lenses for Multimodal Polygonal Prompting

Understanding the translation of polygonal forms into diverse sensory outputs requires examining the problem through multiple theoretical and technological lenses. These lenses provide frameworks for identifying relevant features, grounding translation choices, understanding technological capabilities, and anticipating interaction challenges.

2.1. Cross-Modal Correspondences

A foundational aspect of multimodal translation lies in the phenomenon of cross-modal correspondences, where features of stimuli in one sensory modality are consistently associated with features in another modality by human observers.[6] These associations are often shared across large populations and can influence perception and judgment.[6] Such correspondences can arise from different mechanisms: *structural* correspondences based on neural organization, *statistical* correspondences learned from environmental regularities, or *semantically mediated* correspondences arising from shared language or concepts.[6]

These correspondences offer a psychologically grounded basis for mapping polygon features to sensory parameters:

- **Angularity/Sharpness ↔ Pitch/Timbre/Tactile Sharpness:** Perhaps the most well-known correspondence is the "bouba/kiki" effect (also Takete/Maluma), where rounded shapes are associated with words like "bouba" (containing rounded vowels and sonorant consonants) and spiky, angular shapes with words like "kiki" (containing voiceless stops and high-front vowels).[7] This extends beyond language; people consistently match higher-pitched sounds with more angular shapes and lower-pitched sounds with rounded shapes.[6] This link likely relates to shared concepts of "sharpness"—abrupt changes in a visual contour correspond to abrupt changes or higher frequencies in sound.[9] Tactile sensations of sharpness are also implicitly linked.[6]

- **Size ↔ Pitch/Loudness:** There is evidence that smaller visual objects are associated with higher-pitched sounds, while larger objects are associated with lower-pitched sounds.[6] This may stem from statistical regularities in the environment, where larger objects often produce lower-frequency sounds due to physics.[6] Similarly, larger size can correspond to greater loudness.[6]
- **Complexity/Regularity ↔ Auditory/Olfactory Complexity:** While less directly studied for geometric shapes, the principle that stimulus complexity in one domain might map to complexity in another is plausible. Research in olfaction suggests that the structural complexity of odorant molecules influences the number of perceived olfactory notes and the pleasantness of the smell.[10] Extrapolating, the geometric complexity of a polygon (e.g., number of sides, irregularity, deviation from convexity) could potentially be mapped to auditory complexity (e.g., spectral richness, rhythmic variation, harmonicity) or olfactory complexity (e.g., number of distinct scent notes, faceted nature [11]). Regularity and symmetry, conversely, might map to simpler, more harmonic sounds or balanced scents.

A crucial aspect of these correspondences, particularly relevant for LLM-based systems, is the role of **semantic mediation**. Cross-modal associations are not purely perceptual; they are often intertwined with language and abstract concepts.[12] We use shared descriptors across senses, such as "sharp" sound, "sharp" corner, "sharp" taste, or "warm" color, "warm" sound, "warm" feeling.[9] Conceptual metaphors bridge abstract ideas and concrete sensory experiences (e.g., a "rough" texture and a "rough" day).[14] LLMs, being fundamentally models of language and semantic relationships [16], are uniquely positioned to leverage these connections. Geometric features can be described linguistically (e.g., "a polygon with 5 sharp concave vertices," "a highly symmetric, rounded polygon"). Similarly, target sensory outputs can be described using established vocabularies (e.g., "a high-pitched, dissonant sound," "a smooth, flowing tactile sensation," "a complex fragrance with woody base notes and bright citrus top notes").[18] An LLM could therefore act as a semantic bridge: translating the textual description of a polygon's geometric properties into a textual description of the desired sensory output characteristics. This generated description then serves as a prompt for the modality-specific generative model (e.g., text-to-audio [20], text-to-scent [21]). This semantic pathway is particularly promising for modalities like olfaction, where direct, perceptually robust correspondences with geometry are less established than for sound or touch. The LLM's ability to work with abstract concepts and linguistic metaphors allows it to potentially infer plausible mappings even in the absence of strong, direct perceptual links.

2.2. Computational Geometry & Perceptual Shape Descriptors

To translate polygonal form, we first need robust methods to quantify it. Computational geometry provides algorithms for analyzing geometric objects [22], but standard metrics must be chosen carefully to ensure they align with human perception, which is the ultimate arbiter of the translation's success.

Key polygonal features amenable to quantification include:

- **Edge Count (n):** A fundamental discrete property.[24]
- **Angularity:** Can be measured by the magnitude of interior or exterior angles at each vertex.[24] Sharpness

might relate to the minimum angle, the variance of angles, or metrics capturing abrupt changes in contour direction.[9]

- **Symmetry:** Quantifiable by the number of lines of reflectional symmetry and the order of rotational symmetry (which equals n for regular n-gons).[24] Symmetry is a salient perceptual feature influencing aesthetic judgments and brand personality inferences.[26]
- **Size/Scale:** Measurable via perimeter, area, maximum diameter, or bounding box dimensions.[24]
- **Complexity/Regularity:** Can be assessed by deviations from regularity (equal side lengths and angles define a regular polygon [24]). Other measures might include the number of vertices, concavity, aspect ratio, or metrics derived from computational geometry constructs like the convex hull or triangulation.[22] "Jaggedness" or deviation from smoothness could be relevant descriptors.[22]

Critically, human perception of shape is complex and not always captured by simple geometric measures. Models like **ShapeComp** have been developed to predict perceived shape similarity by integrating over 100 shape descriptors, ranging from simple metrics like area and compactness to more complex ones like Fourier descriptors and shape skeleton analysis.[27] These features are computed from the shape boundary and aim to create a perceptually meaningful multidimensional shape space.[27] Using such perceptually validated descriptors as input for the cross-modal translation process is likely crucial for generating outputs that feel intuitively connected to the source polygon. A full list of ShapeComp features can be found in supplementary materials of the original publication.[27]

The need for explicit, robust feature extraction is further underscored by the documented limitations of current Multimodal Large Language Models (MLLMs) in geometric perception. While MLLMs show remarkable capabilities in high-level visual-linguistic tasks [29], benchmarks like **GePBench** reveal significant deficiencies in their understanding of fundamental geometric properties.[29] GePBench evaluates MLLMs on tasks involving location, size, existence, counting, reference, and relationships of geometric shapes.[29] Even state-of-the-art models like GPT-4o struggle considerably, achieving near-random accuracy on tasks like judging shape size (around 17.5-20% accuracy).[31] Humans, in contrast, perform these tasks almost perfectly.[31] This performance gap strongly suggests that current MLLMs lack the precise internal spatial awareness and geometric reasoning capabilities required for accurately interpreting the specific form features of a polygon directly from an image input. Relying on an MLLM to implicitly "see" and analyze the geometry for translation is therefore unlikely to yield systematic or reliable results. Consequently, a necessary step in the proposed system architecture involves using dedicated computational geometry algorithms or perceptual models (like ShapeComp) to explicitly extract quantifiable features from the input polygon. These structured features then serve as a reliable, unambiguous input to the LLM core, bypassing the MLLM's perceptual weaknesses and ensuring the translation process is grounded in the actual geometric properties of the shape.

2.3. Generative AI Capabilities and Limitations

The feasibility of multimodal polygonal prompting depends heavily on the capabilities of generative AI models for each target sensory modality.

- **Audio Generation:** Models like Meta's AudioCraft suite (including MusicGen for music and AudioGen for sounds) demonstrate the ability to generate high-fidelity audio from text descriptions.[20] These models typically use an encoder (like EnCodec) to learn discrete audio tokens from raw waveforms, an autoregressive language model to predict token sequences based on conditioning input (e.g., text), and a decoder to reconstruct the waveform.[20] Research indicates these models can handle relatively complex and structured text prompts by leveraging techniques like data augmentation (mixing audio and text samples during training), using powerful pre-trained text encoders (like T5), and employing classifier-free guidance to improve adherence to the prompt.[20] This suggests that if polygonal features can be translated into sufficiently descriptive text prompts (potentially by an LLM), current audio models could generate corresponding sounds with control over pitch, timbre, rhythm, and loudness.

- **Tactile Generation:** Mid-air haptic technologies using focused ultrasound allow for generating tactile sensations on the user's skin without physical contact.[3] Techniques involve modulating ultrasound focus points to create sensations like points, lines, and shapes.[4] Research on rendering 2D shapes like circles, squares, and triangles shows that dynamic rendering (moving a tactile point along the perimeter) is significantly more effective for shape recognition than static rendering (presenting the full outline simultaneously).[3] Crucially, introducing pauses at the corners of polygons dramatically improves recognition accuracy, aiding the perception of angularity.[3] Parameters like oscillation frequency, movement speed, and pause duration can be controlled.[3] Other approaches involve vibrotactile actuators embedded in surfaces or wearables. The challenge lies in the precise spatiotemporal control needed to render complex geometric details effectively.
- **Olfactory Generation:** AI-driven scent generation is an emerging field. Systems like Osmo aim to digitize smell by mapping molecular composition to perception (creating a "Primary Odor Map" or POM) and potentially generating novel scents.[5] Other research, like the OGDiffusion model, uses generative diffusion networks trained on mass spectrometry data of essential oils and associated odor descriptors (e.g., "citrus," "woody").[21] Users specify desired scent characteristics, and the AI generates a chemical profile and calculates the required mix of essential oils.[21] Human sensory tests confirm these AI-generated scents align with intended profiles.[21] Current olfactory generation AI primarily relies on semantic descriptors as input, fitting well with an LLM-mediated approach but lacking direct geometric input mechanisms. The technology is less mature than audio or haptic generation, with challenges in fidelity, control, and the complexity of human olfactory perception.[10]

LLM Augmentation: LLMs can serve as the crucial intermediary in this process. Given the explicit geometric features extracted (as per Section 2.2), the LLM can:

1. Interpret these features.
2. Apply knowledge of cross-modal correspondences (Section 2.1) or learned mappings.
3. Translate the geometric information into modality-specific parameters or descriptive prompts.
4. Generate structured input suitable for the respective generative models (e.g., detailed text for AudioGen, path/timing parameters for haptic rendering, scent descriptors for OGDiffusion).[16]

Limitations: Key limitations include the inherent difficulty of translating abstract geometry into certain sensory domains (especially olfaction), the varying degrees of controllability and fidelity in current generative models, the reliance on explicit feature extraction due to MLLM weaknesses [29], and the technological constraints of output hardware.

2.4. Sensory Representation Ontologies

Effective translation requires not only understanding the input (geometry) but also having a structured way to represent the target output (sensory experiences). Taxonomies and ontologies provide formal, semantic models for organizing knowledge about sensory perception.[35]

- **Tactile Ontologies:** Research is actively developing vocabularies and taxonomies to describe tactile perception, often based on psychophysical evaluations of materials.[18] These vocabularies typically distinguish between a *sensory layer* (describing physical properties like soft, smooth, rough, bumpy, moist, elastic) and an *affective layer* (describing subjective responses like pleasant, comfortable, luxury).[18] These structured terms provide potential targets for mapping geometric features. Furthermore, tactile sensations are deeply linked to conceptual metaphors (e.g., weight influencing judgments of importance, texture influencing social judgments) [14], providing another layer of semantic connection that LLMs could potentially exploit. HCI utilizes various haptic feedback systems, implicitly building on these perceptual categories.[37]
- **Olfactory Ontologies:** The fragrance industry traditionally uses the "olfactory pyramid" model to describe perfume structure, consisting of volatile *top notes*, characteristic *heart notes*, and persistent *base notes*.[11] Perfumes are also classified into broad "olfactory families" (e.g., floral, oriental/amber, woody, citrus, chypre, fougère).[11] More detailed descriptions involve "olfactory facets" (e.g., spicy, fruity, green, powdery, animalic) that add complexity.[11] AI scent generation models often work with simpler descriptors like "citrus," "woody," or "floral".[21] Research also links molecular complexity to the perceived number of olfactory notes and pleasantness.[10] Analogies between visual aesthetic principles like complexity and symmetry and fragrance composition or brand perception exist, suggesting shared underlying dimensions of evaluation.[11]

These ontologies and descriptive frameworks are vital for the translation process. They provide a structured target space for the LLM. Instead of mapping geometry directly to low-level signal parameters, the LLM can translate geometric features into high-level sensory descriptors drawn from these established vocabularies (e.g., mapping high angularity to "sharp" auditory timbre and "rough" tactile texture; mapping high symmetry to "balanced" olfactory profile). These descriptors then guide the generative models.

2.5. HCI and Interaction Design

Ultimately, a multimodal polygonal prompting system must be usable and meaningful for humans. HCI principles are essential for designing the interaction.

- **Input Methods:** How do users create or specify the input polygon? Options include direct manipulation drawing tools, parametric input (specifying n, angles, etc.), or uploading existing geometric data or images.
- **Interaction Challenges:** Users, especially novices, often struggle to translate their creative goals into effective prompts for generative AI.[16] There's a "gulf of envisioning" where it's hard to predict the output from the input.[16] For sensory outputs, ensuring perceivability and discriminability is key. Mid-air haptics, for instance, loses the natural exploratory procedure of contour following available with physical touch, making shape identification inherently more difficult unless specific rendering strategies are employed.[3] Generated outputs might be perceived as generic or lacking nuance if the mapping is too simplistic.[16]
- **Potential Solutions:** Structured input interfaces, like the panel-based system in ACAI which separates branding, audience/goals, and inspiration inputs, can help users provide richer context and achieve better alignment compared to simple text prompts.[16] For haptics, dynamic rendering with temporal cues (like pauses at corners) demonstrably improves shape recognition.[3] Providing users with feedback during the generation process and allowing for iterative refinement is crucial for co-creative workflows.[16] Rigorous user studies involving psychophysical testing and qualitative feedback are essential to validate mappings and refine the interaction design.[3]

3. Proposed Translation Strategies and Mapping Framework

Based on the analytical lenses discussed above, specific strategies can be proposed for translating key polygonal features into parameters controlling auditory, tactile, and olfactory generative systems. These strategies prioritize mappings grounded in cross-modal correspondences and HCI findings where available, resorting to semantic mediation and structural analogies for more abstract connections.

3.1. Feature Extraction

As established in Section 2.2, due to the limitations of current MLLMs in geometric perception [29], the first step must be the explicit extraction of quantifiable geometric and perceptual features from the input polygon. This involves analyzing the polygon's vertices and edges to compute:

- Edge Count (n) [24]
- Angularity metrics (e.g., minimum/maximum/variance of interior angles, sharpness measures) [9]
- Symmetry measures (e.g., number of reflectional axes, order of rotational symmetry) [24]
- Size/Scale metrics (e.g., perimeter, area, bounding box) [24]
- Complexity/Regularity metrics (e.g., deviation from regular polygon properties, concavity index, potentially features from models like ShapeComp such as compactness, Fourier descriptors) [24]

This structured feature set forms the input to the LLM core for subsequent translation.

3.2. Auditory Mapping

Leveraging known sound-shape correspondences [6] and sonification principles [38]:

- **Pitch:** Primarily map angularity/sharpness to pitch, with sharper angles corresponding to higher pitches.[6] Map overall size inversely to pitch base frequency (smaller polygons = higher base pitch).[6]
- **Timbre:** Map angularity to timbral brightness or sharpness (e.g., increased high-frequency content, faster attack envelope for more angular shapes).[9] Map geometric complexity/regularity to timbral complexity (e.g., regular polygons = harmonic, simple timbres; irregular/complex polygons = inharmonic, richer, or noisier timbres). Use established timbral descriptors (bright, warm, metallic, rough) as targets.[12]
- **Rhythm/Temporal Structure:** Map edge count (n) to the number of distinct sonic events within a cycle or a rhythmic pattern based on n beats. Use temporal cues like brief pauses, clicks, or changes in timbre to demarcate vertices, especially for conveying angularity, analogous to effective dynamic haptic rendering.[3] Map symmetry to rhythmic regularity, perhaps using palindromic structures for reflectional symmetry. The MIT Sonification Toolkit includes polygon sonification, suggesting established practices exist, although specific algorithms are not detailed in available materials.[40]
- **Loudness:** Map overall size to loudness or dynamic range (larger polygons = potentially louder or wider dynamic range).[6]

Parameter Mapping Sonification (PMSon) provides a general framework where data features are mapped to sound synthesis parameters.[38] While direct examples of polygon feature mapping are scarce in the surveyed PMSon literature [39], the principles of selecting appropriate data features, sound dimensions, and mapping functions apply directly.

3.3. Tactile Mapping

Based on research in mid-air haptics and tactile perception [3]:

- **Rendering Method:** Employ dynamic rendering, tracing the polygon's outline with a moving tactile point or line, as this significantly improves shape recognition over static methods.[3]
- **Corner Representation (Angularity):** Introduce distinct pauses of the tactile point at each vertex. The duration of the pause could potentially correlate with the angle (longer pause for sharper angles?), although optimal fixed durations (e.g., 300ms for square, 467ms for triangle) have been shown effective.[3] Alternatively, use brief, localized vibration pulses of varying intensity/frequency at corners to signal sharpness.
- **Texture/Roughness:** Map geometric complexity or irregularity to the perceived texture of the dynamic trace. For example, a smooth, regular polygon could be rendered with a constant, smooth vibration, while a complex, irregular polygon might use a more varied, perhaps rougher or bumpier vibration pattern, drawing on tactile vocabulary like "smooth," "rough," "bumpy".[18]
- **Outline Speed/Pattern:** The speed at which the tactile point traces the outline could be mapped to the polygon's size (slower for larger polygons?) or complexity. For symmetrical polygons, the tracing pattern itself could be symmetrical.

3.4. Olfactory Mapping (Speculative)

This modality presents the greatest challenge due to the abstract nature of the connection between geometry and smell. Mappings will likely rely heavily on semantic mediation (Section 2.1) and analogies with established olfactory structures and aesthetic principles.

- **Complexity:** Map geometric complexity (e.g., high edge count, irregularity, concavity) to olfactory complexity, interpreted as the number of distinct olfactory notes or the "faceted" nature of the fragrance.[10] Simple, regular polygons (e.g., triangle, square) might map to simpler scents with fewer dominant notes (e.g., a soliflore or a clear citrus). Complex, irregular polygons might map to more multifaceted fragrances with distinct top, heart, and base developments.[19]
- **Angularity/Symmetry → Affective/Structural Qualities:** This requires leveraging semantic links and metaphors.
  - *Angularity:* Could map to descriptors perceived as "sharp," "piercing," "edgy," "stimulating," or "bright" in the olfactory domain (e.g., sharp citrus top notes, pungent spices, green notes).
  - *Symmetry:* Could map to descriptors like "harmonious," "balanced," "rounded," "smooth," or "coherent" (e.g., well-blended florals, smooth woody-amber accords, symmetrical olfactory pyramid structure [19]). This draws an analogy from visual aesthetics where symmetry is associated with harmony and balance.[26]
- **LLM Role:** The LLM translates the extracted geometric features (e.g., "Polygon: N=7, irregular, high angular variance, moderate complexity") into a target scent profile using these speculative mappings and established olfactory language (e.g., "Target Scent: Complex, slightly dissonant, sharp green top notes over a multifaceted woody base"). This descriptive profile then serves as input to an olfactory generative model like OGDiffusion [21] or a system based on Osmo's principles.[5]

3.5. The Mapping Framework (Deliverable 1)

The proposed strategies are consolidated in the table below, providing a conceptual framework for translating polygonal features into multisensory parameters.

Table 1: Proposed Polygon Feature to Sensory Parameter Mappings

Polygonal Feature	Auditory Parameter	Tactile Parameter	Olfactory Parameter (Speculative)	Rationale/Evidence
Edge Count (n)	Number of discrete events/pulses in rhythm; Rhythmic complexity	Number of pauses/pulses during dynamic trace	Potential influence on complexity (higher n → more facets?)	Structural analogy; Temporal segmentation
Angularity/ Sharpness	Higher pitch; Sharper/brighter timbre (more harmonics, fast attack)	Distinct pause/pulse at vertices; Higher intensity/frequency pulse for sharper angles?	"Sharp," "edgy," "bright" descriptors (e.g., citrus, green notes, spices)	Cross-modal correspondence (bouba/kiki) [6]; Semantic link ("sharp") [9]; Haptic findings [3]
Symmetry	Rhythmic regularity; Palindromic structures; Harmonic consonance	Symmetrical tracing pattern/vibration modulation	"Balanced," "harmonious," "smooth" descriptors; Coherent structure	Structural analogy; Aesthetic principles [26]; Olfactory structure [19]
Size/Scale	Lower base pitch for larger size; Louder volume/wider dynamic range for larger size	Slower tracing speed for larger size?; Overall amplitude/area of haptic stimulation (if applicable)	Intensity/Projection? (Larger → stronger scent?)	Cross-modal correspondence [6]; Physical analogy
Complexity/ Regularity	Timbral complexity (inharmonicity, noise); Rhythmic irregularity; Dissonance for irregularity	Vibration texture (smooth for regular, rough/bumpy for irregular); Variability in tracing speed/intensity	Olfactory complexity (number of notes/facets); "Faceted," "complex," "unconventional" descriptors for irregularity	Molecular complexity analogy [10]; Aesthetic principles [26]; Olfactory structure [11]; Tactile vocabulary [18]

Note: Olfactory mappings are highly speculative and rely heavily on semantic mediation and structural analogies.

4. LLM-Augmented System Architecture Proposal (Deliverable 2)

To implement Multimodal Polygonal Prompting, a system architecture integrating geometric analysis, LLM-based translation, and modality-specific generation is proposed.

4.1. Components

The system comprises the following interconnected modules:

1. **Input Module:** Provides the interface for users to define or input the polygon. This could be a graphical drawing tool, a set of parametric sliders (for n, regularity, etc.), an interface to upload vector graphics files (e.g., SVG), or potentially an image analysis component (though direct image input to the LLM for geometry is discouraged, see Section 2.2).
2. **Geometric Feature Extractor:** This crucial module takes the polygon definition from the Input Module and

performs computational analysis to extract the key quantitative geometric and perceptual features outlined in Section 3.1. It utilizes libraries for computational geometry [22] and potentially implements algorithms from perceptual models like ShapeComp.[27] This module outputs a structured representation of the polygon's features (e.g., a JSON object or feature vector). This explicitly addresses the need to bypass MLLM geometric perception weaknesses.[29]

3. **LLM Core (Prompt Augmentation/Translation):** This is the central intelligence hub. It receives the structured geometric features from the Extractor. Leveraging its pre-trained knowledge base, which includes semantic relationships, conceptual metaphors [12], and potentially fine-tuning on cross-modal correspondence data, the LLM performs the core translation task. Based on the Mapping Framework (Table 1), it translates the input geometric features into target sensory characteristics for each modality (auditory, tactile, olfactory). It then generates tailored, structured prompts or sets of control parameters suitable for input into the downstream generative models. Techniques for structured prompting or context integration [16] would be employed here to ensure the generated prompts effectively guide the specialized models.
4. **Modality-Specific Generative Models:** These are specialized AI models responsible for synthesizing the sensory output based on the instructions from the LLM Core:
   - **Audio Generator:** A text-to-audio or music generation model (e.g., based on AudioCraft architecture [20]) that takes the LLM-generated textual description or parameter set and synthesizes the corresponding sound waveform.
   - **Haptic Renderer:** A control system that takes path, timing, intensity, frequency, and pause parameters from the LLM and drives a haptic output device (e.g., an ultrasonic phased array for mid-air haptics [3] or a matrix of vibrotactile actuators).
   - **Olfactory Synthesizer:** An AI model (e.g., based on OGDiffusion principles [21] or Osmo's mapping approach [5]) that takes the LLM-generated scent descriptors or target chemical profile and computes an essential oil recipe or controls a scent generation/diffusion mechanism.
5. **Output Transducers:** The physical hardware that delivers the sensory stimuli to the user:
   - Audio: Speakers or headphones.
   - Tactile: Mid-air haptic device, wearable vibrotactile array, or surface haptics.
   - Olfactory: Computer-controlled scent diffuser or olfactometer.
6. **User Feedback/Refinement Loop:** An interface allowing users to perceive the generated multisensory output and provide feedback or make adjustments. This feedback could be used to refine the parameters directly or potentially fed back to the LLM Core to improve subsequent translations, enabling a co-creative interaction.[16]

4.2. Information Flow

The typical flow of information through the system is as follows:
User Input (Polygon Definition) → Input Module → Geometric Feature Extractor → Structured Geometric Features → LLM Core → (Generates Modality-Specific Prompts/Parameters) → ``
→ Sensory Signals → Output Transducers → Multisensory Output to User → (Optional) User Feedback → Refinement Loop (adjusting parameters or LLM guidance).

4.3. Role of the LLM

It is critical to emphasize the LLM's role in this architecture. It does not perform the raw sensory signal generation. Instead, it acts as an **intelligent translator and orchestrator**. It leverages its vast semantic knowledge and understanding of relationships (including cross-modal ones) to bridge the abstract domain of geometry with the concrete domains of sound, touch, and smell. By operating on explicit, structured geometric features and generating structured prompts or parameters for specialized generative models, the architecture plays to the strengths of LLMs (semantic reasoning, translation) while mitigating their weaknesses (geometric perception, direct signal synthesis).

5. Evaluation Metrics and Challenges (Deliverable 3)

Evaluating the success of a Multimodal Polygonal Prompting system is complex, requiring metrics that capture not only the technical fidelity of the translation but also the quality of the human perceptual experience.

5.1. Defining Success

Success can be defined along several dimensions:

- **Fidelity:** Does the generated sensory output accurately and consistently reflect the specific geometric features of the input polygon according to the intended mapping? For example, does increasing the angularity of the input polygon reliably lead to a perceived increase in auditory pitch or tactile sharpness?
- **Discriminability:** Can users reliably distinguish between the sensory outputs generated from different input polygons? For instance, can a user tell the difference between the haptic rendering of a square and a triangle [3], or the sonification of a pentagon versus a hexagon?
- **Interpretability:** Do users understand the underlying mapping between geometric features and sensory parameters? Can they predict the sensory output given a polygon, or infer the polygon's properties from the sensory output? Is the translation intuitive?
- **Affective Response/Engagement:** Beyond accuracy, is the resulting multisensory experience aesthetically pleasing, engaging, informative, or useful within a specific application context? Does it evoke the intended emotional or qualitative response (e.g., does a symmetrical polygon generate a scent perceived as "harmonious" [26])?

5.2. Proposed Metrics

A combination of quantitative and qualitative methods is necessary for comprehensive evaluation:

- **Psychophysical Tests:** These are essential for measuring perceptual fidelity and discriminability. Examples include:
  - *Matching Tasks:* Presenting a polygon visually and asking users to select the corresponding sound, tactile stimulus, or scent from a set of options.
  - *Identification Tasks:* Presenting only the sensory stimulus (e.g., haptic shape trace) and asking users to identify the corresponding polygon shape (e.g., circle, square, triangle).[3] Accuracy and confidence ratings are key metrics.
  - *Discrimination Tasks:* Presenting pairs of sensory stimuli generated from slightly different polygons and asking users if they are the same or different (measuring Just Noticeable Differences).
  - *Rating Scales:* Using Likert scales or visual analog scales to rate perceived qualities of the sensory output (e.g., perceived pitch, sharpness, roughness, complexity, pleasantness) based on established sensory vocabularies.[10]
- **Information Theory Measures:** Techniques like mutual information analysis can quantify how much information about specific input polygon features (e.g., edge count, symmetry type) is successfully transmitted through each sensory output channel.
- **Qualitative Feedback:** User interviews, questionnaires, and think-aloud protocols during interaction can provide rich insights into the subjective experience, interpretation strategies, intuitiveness of the mapping, and overall engagement.
- **Task-Based Evaluation:** If the system is designed for a specific application (e.g., data exploration,

accessibility aid), evaluate user performance on relevant tasks (e.g., speed and accuracy in identifying geometric patterns in data represented multimodally).

5.3. Key Challenges

Evaluating multimodal polygonal prompting faces several inherent challenges:

- **Subjectivity:** Sensory perception, especially olfaction and affective responses to any sense, is highly subjective and influenced by individual factors like experience, genetics, culture, and context.[19] What constitutes a "good" or "accurate" translation can vary significantly between users.
- **Perceptual Limits:** Human sensory systems have limits in resolution and discrimination ability. Fine-grained geometric differences may not be perceivable when translated into certain sensory parameters, particularly for smell, which is often described as ineffable.[12]
- **Lack of Ground Truth:** Unlike tasks with objective correct answers, defining the single "correct" sound, touch, or smell for a given polygon is impossible. Evaluation relies on consistency with chosen mappings (often based on correspondences) and perceptual coherence, rather than absolute correctness.
- **Technological Limitations:** The fidelity, controllability, and range of current generative models and output hardware impose constraints. Mid-air haptics has resolution limits [3], audio models might struggle with highly specific timbral requests, and olfactory generation faces challenges in rapid switching, intensity control, and recipe accuracy.[10]
- **Mapping Complexity:** Translating multiple interacting geometric features (e.g., simultaneous changes in edge count, angularity, and symmetry) into multiple interacting sensory parameters across three modalities presents a significant design challenge. Avoiding sensory overload and ensuring the mapping remains interpretable requires careful balancing.

6. Reflexive Insights and Future Directions (Deliverable 4)

The investigation into Multimodal Polygonal Prompting reveals several deeper implications and highlights critical areas for future work.

6.1. Strengths and Limitations of the Approach

The proposed approach, centered on explicit feature extraction followed by LLM-mediated translation, holds several strengths. It leverages established cross-modal research to ground mappings psychologically.[6] It utilizes the semantic power of LLMs to bridge abstract geometry and concrete sensation, particularly for challenging modalities like olfaction.[12] By incorporating explicit geometric feature extraction, it directly addresses the documented weaknesses of current MLLMs in geometric perception, ensuring the translation process is driven by accurate geometric data.[29]

However, the approach also has limitations. It relies heavily on the accuracy and perceptual relevance of the extracted geometric features (e.g., from ShapeComp [27]). The proposed mappings, especially for olfaction, remain speculative and require significant empirical validation. The overall system complexity is high, involving multiple stages of analysis and generation. Furthermore, the entire endeavor depends on the continued rapid advancement of generative AI capabilities and sensory output technologies.

6.2. The "Ontological Scaffolding" Hypothesis

Engaging with abstract geometric forms through multiple sensory modalities simultaneously could have interesting cognitive implications. Theories of embodied cognition suggest that our conceptual knowledge is grounded in sensory and motor experiences.[14] Physical touch experiences, for instance, may create an "ontological scaffold" for developing abstract concepts related to weight, texture, or hardness, which are then applied metaphorically to non-physical domains.[14] Experiencing the "sharpness" of a triangle not just visually, but also through a corresponding high-pitched, sharp sound and a distinct tactile pulse at the corners, might reinforce the abstract concept of "sharpness" itself. A system for multimodal polygonal prompting could serve as a platform to investigate these ideas, exploring how curated multisensory experiences related to geometric forms influence the learning and understanding of abstract spatial and mathematical concepts.

6.3. Potential for AI to Discover Novel Cross-Modal Correspondences

While the current proposal relies primarily on known, human-observed cross-modal correspondences [6] as the basis for mapping, the LLM-augmented system itself presents an intriguing possibility for discovering *new* correspondences. AI systems, particularly deep learning models, excel at identifying complex patterns and statistical regularities in large datasets.[1] If the system incorporates a feedback loop where user responses (e.g., discriminability ratings, preference scores, task performance) are used to evaluate the effectiveness of different mappings, the LLM could potentially learn optimized translation strategies. Instead of merely applying predefined rules based on human intuition or existing research, the system could adapt its mappings to maximize perceptual clarity or user satisfaction based on empirical data. This data-driven optimization process might uncover statistically significant relationships between geometric features and sensory parameters that are not intuitively obvious or previously documented in psychophysical literature. For example, perhaps a specific combination of polygon irregularity and concavity consistently maps to a particular olfactory note combination that enhances discriminability, even if the semantic link is unclear. Such discoveries could lead to novel, potentially more effective (albeit perhaps less immediately intuitive) methods for sensory data representation and could even contribute new insights to the scientific understanding of cross-modal perception.

6.4. Ethical Considerations

The development of technologies that translate information across sensory modalities, especially using powerful AI, raises ethical considerations. The potential for using cross-modal effects in advertising or marketing to influence consumer perception and choice requires careful scrutiny.[13] Ensuring that systems designed for accessibility are genuinely helpful and do not impose additional cognitive load is paramount. Managing user expectations regarding the fidelity and capability of nascent technologies, particularly in olfaction and haptics, is important to avoid disappointment or misinterpretation. Transparency about the mapping principles and the limitations of the system is crucial.

6.5. Future Research

This investigation highlights numerous avenues for future research:

- **Empirical Validation:** Rigorous psychophysical studies are needed to validate the proposed mappings, especially the more speculative ones for olfaction and complex feature combinations.
- **Generative Model Development:** Continued progress is required in creating generative models for audio, haptics, and olfaction that offer finer-grained control, higher fidelity, and better handling of structured, parameter-based input.
- **Context and User Goals:** Research should explore how the optimal mapping might change depending on the user's task, goals, and the context of interaction.
- **Dynamic and 3D Forms:** Extending the concept beyond static 2D polygons to dynamic shapes (changing over time) and 3D geometric objects presents significant additional complexity.
- **Evaluation Methodologies:** Developing standardized benchmarks and evaluation protocols for multimodal

translation systems is crucial for comparing different approaches.

- **Improving MLLM Geometric Perception:** Continued research, potentially informed by benchmarks like GePBench [29], is needed to enhance the intrinsic geometric understanding capabilities of MLLMs, potentially reducing the reliance on external feature extractors in the long term.
- **Exploring AI-Driven Discovery:** Systematically investigating the potential for AI to learn and discover novel, effective cross-modal mappings based on user feedback or objective performance metrics.

7. Conclusion

Multimodal Polygonal Prompting presents a fascinating frontier at the intersection of geometry, perception, and artificial intelligence. This report has explored the conceptual underpinnings and practical considerations involved in translating the abstract form features of polygons into auditory, tactile, and olfactory experiences using LLM-augmented generative systems. The analysis indicates that while significant challenges exist, particularly concerning the limitations of current MLLM geometric perception and the nascent state of olfactory generation technology, the approach holds considerable promise.

Leveraging known cross-modal correspondences provides a psychological basis for translation, while the semantic capabilities of LLMs offer a powerful mechanism for bridging modalities, especially through shared language and conceptual metaphors. The proposed architecture, emphasizing explicit geometric feature extraction prior to LLM translation, offers a pragmatic pathway to mitigate current AI limitations. The potential applications in data representation, accessibility, generative art, and HCI are compelling, motivating further research and development.

Key challenges remain in validating mappings empirically, refining generative technologies, establishing robust evaluation methods, and navigating the complexities of subjective sensory perception. However, the exploration of this domain not only pushes the boundaries of generative AI but also offers unique opportunities to probe the nature of human multisensory integration and potentially uncover novel relationships between abstract form and concrete sensation. The continued development of multimodal AI systems, guided by principles from perception science and HCI, promises to unlock rich and previously inaccessible ways of interacting with and understanding information.

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